

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

TypeScript, JavaScript, Python, SQL, C++, Java, Bash, HTML/CSS

LLM & Agentic Systems

LLM agents, tool use, function calling, MCP, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama

Frontend & Full-Stack

React, Electron, Webpack, GraphQL, REST APIs, integration testing, design patterns

Backend & Data

FastAPI, Node.js, Express, Fastify, async services, MongoDB, PostgreSQL, Redis, SQLite, sentence-transformers, RAG

Cloud & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Took a full-stack **internal product** (React, Node.js, MongoDB) from **zero to one** for Wysa's AI team to ingest, label, and centralize training data; replaced spreadsheet workflows with secure role-based REST APIs and reviewer UIs, cutting manual operations by **100+ hours/month** and lifting labeling throughput **60%**.
- Owned **multi-tenant NHS eTriage backend** (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; hardened auth, rate limiting, and remediated **20+ VAPT findings** (contributed to **\$340K+** in revenue).
- Acted as primary technical point of contact for the eTriage backend, partnering directly with clinical, AI, and product stakeholders to scope and ship features (auth, rate limiting, audit logging), and ran knowledge-transfer sessions for new developers joining the UK clinical team.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped **LLM-powered context-aware replies** into a customer-facing chat product, iterating on prompt design and response-quality heuristics from real usage; designed and implemented **20+ production RESTful APIs** in Node.js + Fastify over PostgreSQL with clear contracts and **JS→TS migration** across multiple repositories for long-term maintainability.

Projects

cf-ai-research-scout - Full-Stack TypeScript AI Agent on Cloudflare

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an **AI agent** end-to-end with a TypeScript frontend that fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time, prototyped on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries via WorkflowEntrypoint and Durable Objects so multi-step tasks survive process restarts and tool failures, with agent state and conversation history persisted in D1 without an external database.

AgenticSearch - Agentic LLM Service Built FOR Language Models

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic service that converts open-ended topic queries into structured entity tables via typed retrieval planning, page rerank, deterministic parsers with **LLM fallback**, output verification, and cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, **provider routing/fallback across OpenAI and Groq**, pipeline instrumentation, and a reviewer-facing trust UI that surfaces extraction quality, source URLs, and ranking transparency to the people using the system.

Sonare - Zero-to-One Full-Stack Desktop Product

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Took a **cross-platform desktop product** from zero to one under hackathon time pressure: React/Electron frontend, FastAPI backend, on-device pipeline integrating MediaPipe hand tracking, whisper.cpp Automatic Speech Recognition (ASR), and local Text-To-Speech (TTS).
- Engineered the **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling video and audio streams locally on device.

Publication & Extracurriculars

- **Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.